

SAP ABAP Production Issue #001

Excel Attachment Date Changes from MM/DD/YYYY to DD.MM.YYYY While Sending Email

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Difficulty Level: Intermediate

Introduction

One of the most common issues faced by SAP ABAP developers occurs when generating Excel attachments and sending them through email.

The ABAP program displays dates correctly inside SAP, but once the user opens the Excel attachment, the dates appear in a completely different format.

For example:

Expected:

28.06.2026

Actual:

06/28/2026

This issue often causes confusion for business users, especially in multinational organizations where different regional settings are used.

Business Scenario

A business process generates an agreement summary report.

The report is converted into an Excel attachment and automatically emailed to business users every morning through a background job.

The business requires all dates to appear in the following format:

DD.MM.YYYY

Example:

Agreement Date

28.06.2026

15.07.2026

01.01.2027

However, after opening the Excel attachment, users observed:

Agreement Date

06/28/2026

07/15/2026

01/01/2027

The data stored in SAP was correct.

The issue only appeared after opening the Excel file.

Initial Analysis

The first assumption was that SAP was exporting the date incorrectly.

The internal table contained values of data type **DATS**.

Example:

20260628

During ALV display inside SAP, the output appeared correctly according to the user's logon settings.

Therefore, the issue was not with SAP data storage.

Root Cause

The actual problem was caused by Microsoft Excel.

When SAP exports a field with data type **DATS**, Excel interprets the value according to its own regional settings.

For example:

A user in the United States may see:

06/28/2026

A user in Germany may see:

28.06.2026

A user in the United Kingdom may see:

28/06/2026

The same Excel file can therefore display different date formats on different computers.

SAP does not control how Excel formats dates after the file is opened.

Common Failed Approaches

Attempt 1

Using Conversion Exit

CONVERSION_EXIT_PDATE_OUTPUT

Result

✗ Failed

Reason:

Conversion exits affect ABAP output formatting but do not force Excel to preserve a specific display format.

Attempt 2

Using WRITE ... TO

Example:

WRITE lv_date TO lv_char.

Result

✗ Failed

Reason:

Only suitable when exporting character fields.

Many production reports reuse the same structure for ALV display, processing, and Excel generation, making this approach impractical.

Attempt 3

Changing DATS field to CHAR

Result

✗ Rejected

Reason:

The same internal table was used throughout the application.

Changing the data type introduced side effects in other program logic.

Attempt 4

Applying MASK

Example:

MASK = '____.____.____'

Result

 Failed

Reason:

MASK controls display within ABAP but does not influence Excel's formatting behavior.

Why the Problem Happens

The export process preserves the date value, but Excel automatically converts the cell into a date type and formats it using the operating system's regional settings.

This behavior is outside SAP's control.

Recommended Solution

The safest approach is to generate a dedicated export structure in which only the date fields are converted to formatted character strings before the Excel file is created.

Example:

```
DATA lv_date TYPE char10.
```

```
WRITE ls_data-created_on TO lv_date DD/MM/YYYY.
```

```
REPLACE ALL OCCURRENCES OF '/' IN lv_date WITH '.'.
```

```
ls_export-created_on = lv_date.
```

The export structure now contains:

28.06.2026

instead of a DATS value.

Excel treats the value as text rather than a date, preserving the required format.

Note: The exact implementation depends on how the Excel file is generated (e.g., ALV export, XML Spreadsheet, CSV, or custom XLSX generation). The key principle is that the exported value should be text if a fixed visual format is required.

Production Considerations

Before implementing this solution, consider the following:

- Will users need to sort or filter dates as actual dates in Excel?
- Is the export consumed by another system or only by humans?
- Is the same structure reused elsewhere in the application?
- Are there multiple date fields that require consistent formatting?

If users need Excel date calculations, exporting as text may not be the best option. In such cases, applying an Excel cell style that specifies the date format is preferable, if your export technology supports it.

Testing Checklist

The solution should be tested under the following conditions:

- ✓ Foreground execution
 - ✓ Background execution
 - ✓ Email attachment
 - ✓ Multiple SAP users
 - ✓ Different Windows regional settings
 - ✓ Different Microsoft Excel versions
 - ✓ Large datasets
 - ✓ Empty date fields
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Best Practices

- Keep processing structures separate from export structures when formatting requirements differ.
 - Avoid changing the original DATS field solely for reporting purposes.
 - Document regional formatting requirements with the business team.
 - Test exports on systems configured with different locale settings.
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Common Interview Questions

Why does Excel display different date formats for the same exported file?

Because Excel formats date cells according to the user's regional settings unless a specific cell format or text value is supplied.

Why shouldn't we simply change the DATS field to CHAR?

Doing so can affect other program logic, validations, calculations, and database operations if the structure is shared.

When should dates be exported as text?

When the business requires a fixed visual representation that must remain identical across different user locales and no Excel date calculations are needed.

Lessons Learned

The issue was not caused by SAP.

The issue was caused by Excel interpreting date values according to the user's local settings.

The correct solution depends on the business requirement:

- If a fixed display format is required, export formatted text.
- If Excel date functionality is required, export real dates and apply an appropriate Excel date format when possible.

Understanding the distinction between **data storage** and **presentation** is essential when generating Excel reports from SAP ABAP.

Key Takeaways

- SAP stores dates independently of display format.
- Excel decides how to display date values unless formatting is explicitly controlled.
- Separate export formatting from business logic whenever practical.
- Always validate Excel exports under the same conditions in which end users will open them.